Introduction to Derivatives

BUSS386. Futures and Options

Prof. Ji-Woong Chung

Lecture Outline

- What a derivative is, and the four flavors (forward, future, swap, option)
- Where they trade, how big the market is, why they grew so fast
- What they are good for — and how they blow up
- Probability toolkit: returns, Normal, three risk measures (SD, VaR, ES)

What is a derivative?

- A **derivative** is a financial contract whose value depends on an *underlying variable*.
- *Underlying* = price of a traded asset (equity, bond, FX, commodity), a property of a price (volatility), an event (default), or a measurable observable (weather, inflation).
- Examples:
  - A contract to buy 1,000 barrels of crude oil on Sept 16 next year at \$80/barrel.
  - A contract giving the right (but not obligation) to buy 100 KOSPI 200 ETF shares at ₩42,000 in three months.

Four flavors

- **Forward —**
- **Futures —**
- **Swap —**
- **Option —**

Two families: contract vs. securitized

Contract derivatives

- Forwards, futures, swaps, options.
- Two counterparties, future cash flows.
- Zero-sum: one's gain = other's loss.
- **Transfer** risk.

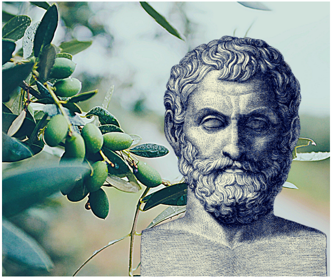
Securitized products

- Pool of assets repackaged into new securities (ABS, MBS, CMO).
- Sliced into **tranches** by payment priority.
- Senior takes AAA + low yield; equity takes first loss + high yield.
- **Redistribute** risk.

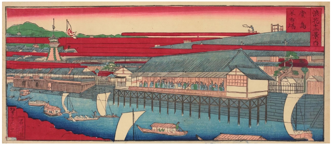
This course focuses on contract derivatives.

A short history — pre-modern to modern

- **600 BC** — Thales buys options on olive presses.
- **1630s** — Amsterdam tulip mania: forwards *and* options on bulbs.
- **1730** — Dojima rice exchange: world's first organized futures market.
- **1848** — Chicago Board of Trade (CBOT): grain futures.
- **1919** — Chicago Mercantile Exchange (CME).
- **1973** — CBOE lists first listed equity options; same year, Black-Scholes published.
- **1996/7** — KRX launches KOSPI 200 futures, then options.



First options: Thales of Miletus.



Dojima Rice Exchange.

Two habitats: exchange vs. OTC

Exchange-traded

- Standardized contracts.
- Central counterparty (CCP) clears every trade.
- Daily margining; transparent prices.
- Futures, listed options.
- Examples: CME, CBOE, KRX.

Over-the-counter (OTC)

- Bilateral, customizable terms.
- Governed by ISDA Master Agreement.
- Counterparty credit risk.
- Forwards, swaps, exotics.
- Players: banks, hedge funds, corporates.

2008 and after: from bilateral to cleared

- **Pre-2008:** OTC derivatives largely unregulated. Banks acted as market makers; trades governed by ISDA Master.
- **Sept 2008 Lehman.** 8,000 counterparties, hundreds of thousands of open derivatives trades. Unwinding takes years.
- **Post-2008 reforms:** (i) standardized OTC contracts must clear through **CCPs**; (ii) all trades reported to a central repository; (iii) collateral required.

Without a CCP, every OTC trade is a private bilateral exposure — one default cascades. With a CCP, the exchange sits between every buyer and seller, collects margin daily, and absorbs a default before it spreads.

How big is this market?

Four ways to measure market size

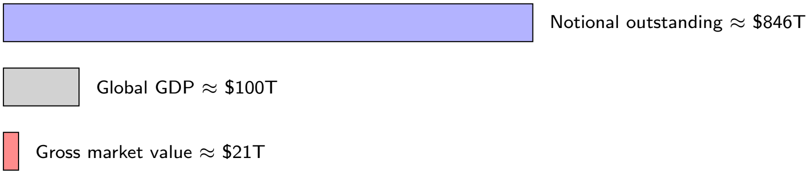
- **Trading volume —**

- **Market value —**

- **Notional value —**

- **Open interest —**

“How big is the derivatives market?”



Data: BIS OTC derivatives statistics, Tables D5.1–D5.2, end-June 2025

https://www.bis.org/statistics/derstats.htm.S

Korea in one slide

- **KRX** lists: KOSPI 200 index futures/options, KTB (Korean Treasury Bond) futures, USD/KRW futures.
- KOSPI 200 options were the **world's most-traded derivative by contract count** from ∼2001 to early 2010s. Retail participation unusually heavy.[1]
- 2011+ regulatory tightening reshaped the market:
  - **Contract multiplier** raised ⇒ small accounts priced out.
  - **Suitability rules** ⇒ brokers must verify knowledge, experience, capital.

Data: KRX `https://data.krx.co.kr/` FSC `https://www.fsc.go.kr/`

[1] The KOSPI200 options accounted for 43.4% of the global trading volume in equity index futures and options in 2011.

Uses of derivatives

- **Risk management.** Hedge an exposure. *Farmer enters a contract that pays when corn is low.* Insurance is a derivative on the loss.
- **Speculation.** Express a leveraged view. *Bet that S&P 500 will be between 1,300 and 1,400 in 1 year* — a tailored, high-leverage payoff.
- **Reduced transaction costs.** *Mutual fund manager wants to rotate from stocks to bonds* — trading derivatives may be cheaper than actually selling & buying.
- **Price discovery.** Futures prices aggregate the market's best guess about the future. *Fed and OPEC both watch oil futures curves.*
- **Regulatory arbitrage.** Circumvent restrictions, taxes, accounting rules. *Synthetic sale of stock* (economically sell but retain physical ownership) to defer tax or keep voting rights.

Three blow-ups, three lessons

- **Barings Bank (1995).** Unauthorized Nikkei 225 futures bets sink a 233-year-old bank.
  $\Rightarrow$ *Instrument was fine; operating model was broken.*
- **Korean SMEs — KIKO (2008–09).** "Cheap" FX options become leveraged short-USD bets when KRW falls; $\sim$\$3–5B SME losses.
  $\Rightarrow$ *Hedge in one tail, leveraged bet in the other.*
- **UK LDI funds (Sept 2022).** 3–7$\times$-leveraged gilt+swap hedges; gilt sell-off triggers margin calls $\rightarrow$ forced selling; BoE intervenes with up to £65B.
  $\Rightarrow$ *Hedge was right; leverage smuggled in liquidity risk.*

Common thread: either the hedge drifts away from the exposure, or leverage turns a price move into a forced sale.

Refs: BoE FSR Oct 2022 https://www.bankofengland.co.uk/financial-stability-report FSS https://www.fss.or.kr/

Statistics: Review

Why we need a probability toolkit

- Every derivative payoff depends on a **future** price we do not yet know.
- To price it we need the *distribution* of that price, not just an average.
- Today's path: random variables $\rightarrow$ Normal $\rightarrow$ three risk measures (SD, VaR, ES).

Simple vs. log returns

- **Simple**: $R_t = \dfrac{P_t - P_{t-1}}{P_{t-1}}$ **Log**: $r_t = \ln\left(\dfrac{P_t}{P_{t-1}}\right)$
- Small moves: $r_t \approx R_t$. Big moves diverge: -10% simple $\rightarrow -10.5\%$ log; -50% simple $\rightarrow -69\%$ log.

Why log returns in this course

(i) **Additive over time**: two-day log return $=$ day-1 $+$ day-2.
(ii) Enables $\sqrt{T}$ vol scaling.
(iii) Black-Scholes (Lec 10) assumes log-normal prices.

Discrete random variables

Example. You buy KOSPI 200 ETF today. What return over the next year?

Scenario	Return r_i	Prob. p_i
Bull market	$+25\%$	0.30
Base case	$+8\%$	0.50
Bear market	-20%	0.20

- **Expected return**:
  $E(R) = \sum_i p_i r_i = 0.30(0.25) + 0.50(0.08) + 0.20(-0.20) = \boxed{7.5\%}$.
- **Variance**: $\mathrm{Var}(R) = \sum_i p_i (r_i - E(R))^2 = 0.0243$.
- **Volatility**: $\sigma(R) = \sqrt{\mathrm{Var}(R)} = \boxed{15.6\%}$.

Read as: "a typical year is 7.5%, but easily $\pm$15.6% around that."

Continuous random variables & the Normal

- Real returns are **continuous**.
- Replace the probability table with a density $f(r)$. Probabilities become areas: $\text{Prob}(R \leq 0.05) = \int_{-\infty}^{0.05} f(r)\,dr$.
- **Normal**: $R \sim N(\mu, \sigma^2)$, density $f(r) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(r-\mu)^2/2\sigma^2}$.
- Linear transforms stay Normal: $aR + b \sim N(a\mu + b,\, a^2\sigma^2)$.

Finance anchor

S&P 500 daily returns: $\approx N(\mu \approx 0.04\%,\ \sigma \approx 1\%)$.
KOSPI 200 daily returns: $\approx N(\mu \approx 0.03\%,\ \sigma \approx 1.2\%)$.

Standard normal & the Z-score

- **Standard normal**: $Z \sim N(0,1)$, CDF $\Phi(x) = \text{Prob}(Z \leq x)$. Excel: `=NORM.S.DIST(x, TRUE)`.
- If $R \sim N(\mu, \sigma^2)$, then

$$Z = \frac{R - \mu}{\sigma} \sim N(0,1).$$

  One table suffices: subtract μ, divide by σ, look up.
- For full Normal: Excel `=NORM.DIST(r, mu, sigma, TRUE)`.

Warm-ups:

Ex. 1 $R_1 \sim N(0,1)$. Find $\text{Prob}(R_1 > 1)$.

Ex. 2 $R_2 \sim N(0.10,\, 0.20^2)$. Find $\text{Prob}(R_2 \leq 0.50)$.

Risk Measures

Standard deviation, volatility, and $\sqrt{T}$ scaling

- **Standard deviation** measures dispersion of outcomes. Larger $\sigma \Rightarrow$ riskier.
- **Disadvantage**: cannot distinguish upside from downside moves.
- **Volatility** $\sigma =$ std. dev. of returns. Time unit matters (daily vs. annual).
- If daily log returns are i.i.d. with daily σ_{daily}:

$$\sigma_{T\text{-day}} = \sigma_{\text{daily}}\sqrt{T} \quad \Longrightarrow \quad \sigma_{\text{annual}} = \sigma_{\text{daily}}\sqrt{252}.$$

- Rule of thumb: **S&P 500** annualized vol $\approx$ 15–20% in calm years, 30%+ in crises. **KOSPI 200** runs slightly higher.

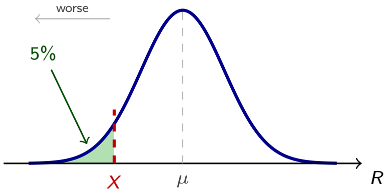
Value at Risk — definition

In one sentence

1-day 95% VaR is the loss level such that, on a typical day, you expect to lose *more* than this only 5% of the time.

Find X such that

$$\text{Prob}(R \leq X) = 0.05$$



- Three knobs: **horizon**, **confidence**, **portfolio value** V.
- VaR scales with V: doubling the position doubles the VaR.
- Excel: `=NORM.INV`$(0.05, \mu, \sigma)$.

Read: "the green 5% tail starts at X. Anything left of X is a worse-than-VaR day."

VaR — examples

Q1 (single stock). Annual return $R \sim N(\mu = 15\%, \sigma = 30\%)$. Find 5% VaR.

Q2 (S&P 500 ETF). \$10M position, daily $\mu \approx 0$, $\sigma \approx 1\%$ ($=$ \$100k). Find 1-day and 10-day 99% VaR.

VaR — 2-stock portfolio

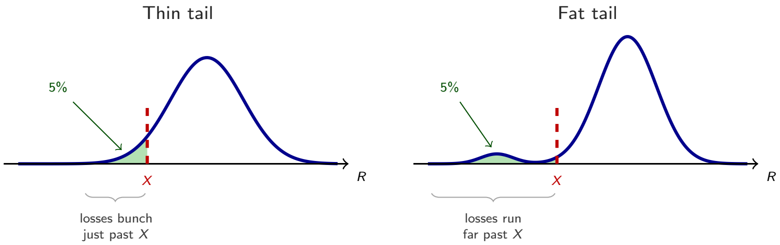
Q. \$5M in A, \$5M in B (\$10M total). Annual:
$\mu_A = 15\%, \sigma_A = 30\%; \mu_B = 18\%, \sigma_B = 45\%; \rho = 0.4$. Find 90% VaR.

NB: if $X \sim N(\mu_x, \sigma_x^2)$, $Y \sim N(\mu_y, \sigma_y^2)$, then $X + Y \sim N(\mu_x + \mu_y,\ \sigma_x^2 + \sigma_y^2 + 2\rho\sigma_x\sigma_y)$.

VaR — historical data & known issues

Historical VaR. Drop the Normal assumption. Use the 5th percentile of past-year daily returns as the 5% VaR estimate.

- **Issue 1.** Assumes future return distribution = past. Crises break this assumption.
- **Issue 2.** VaR is the *minimum* loss at a given probability; it says *nothing about how bad the loss could be* beyond.



Same 5% VaR, very different tails. Need a measure of the tail itself.

Expected Shortfall

- **Expected Shortfall (ES)** answers: "if things get bad, what is the expected loss?"
- Let V be the VaR at probability p (e.g., $p = 5\%$). Then

$$\mathrm{ES}_p = E[\, R \mid R \le V \,].$$

- Historical data: in Excel use `=AVERAGEIF(returns, "<="&V)`.
- **Ex.** Using 1 year of daily returns, compute ES at 5% and 10% (worksheet exercise).
- **Why ES?** It captures *tail size*, not just the threshold. ES is mathematically "coherent"; VaR is not.

Where VaR & ES are actually used

- **Bank capital (Basel)** — regulatory capital tied to VaR/ES of the trading book.
  - 1996: capital $= k \times \text{VaR}(99\%, 10\text{d}),\ \ k \geq 3$.
  - Basel III/IV: shifts to **97.5% Expected Shortfall** for trading book.
- **Asset managers & pensions** — internal risk limits; daily 95%/99% VaR.
- **Exchange margin (KRX, CME)** — initial margin sized to cover a 99%+ daily loss.
- **Insurance (Solvency II, K-ICS)** — 99.5% VaR over a 1-year horizon to size required capital.

Refs: BIS Basel https://www.bis.org/basel_framework/ K-ICS (FSS) https://www.fss.or.kr/

Summary — what to take away

- A derivative is a **contract written on an underlying**. Four flavors — **forward, futures, swap, option** — and everything else this semester is built out of them.
- **Where** it trades sets the credit story: exchange means standardized, margined, cleared; OTC means bespoke and bilateral (and cleared too, for standard swaps, since 2008).
- Derivatives **transfer** risk between counterparties; they do not make it disappear. One side's gain is the other side's loss.
- Barings, KIKO, LDI: the instrument was rarely the problem. **Leverage and mismatch** were.
- Risk vocabulary: **SD** (a typical move), **VaR** (the threshold loss), **ES** (the average loss once you are past it). Basel III/IV runs on 97.5% ES.

Next week: interest rates and discounting — the machinery behind every price we compute.